

Pulsed Nuclear Magnetic Resonance

PHYS3117 - Physics Labotory

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1 Introduction

Nuclear magnetic resonance (NMR) is when a nuclei is placed in a strong constant magnetic field but then a weaker oscillating magnetic field is applied. The nuclei then produces an electromagnetic signal with a frequency characteristic to the magnetic field at the nucleus. When the oscillation of the applied magnetic field is near the intrinsic oscillation frequency of the nuclei, resonance is achieved. The electromagnetic signal can then be read and thus the structure of the matter can be determined.

The most prominent use of NMR technology is in medicine. The magnetic resonance imaging or MRI machine is one of the most important inventions in the last 100 years, advancing the field and the health care system significantly. Generally, NMR spectroscopy is used everywhere to determine physical, chemical, electronic and structural information about nuclear spins in various samples.

2 Theory

Atomic nuclei carry protons and neutrons, which both have a property known as spin angular momentum. Nuclei with non-zero spin possess a magnetic dipole moment, μ , which is proportional to the spin angular momentum, S ,

$$\vec{\mu} = \gamma \vec{S}, \quad (1)$$

where γ is the gyromagnetic ratio, which is a constant that is unique to specific nuclei.

2.1 Larmor Frequency

If we place this magnetic moment in a magnetic field, $\mathbf{B}$, it will experience a torque, τ equal to $\mu \times \mathbf{B}$, as the field rotates the dipole to be aligned. But the magnetic moment has an angular momentum $\vec{S}$ directed perpendicular to the field, causing the magnetic dipole to precess around the direction of the magnetic field. The magnetic moment is now at an angle of θ to the magnetic field, known as the angle of precession. The torque from the magnetic field will change the direction of the angular momentum of the magnetic moment so in a small time t so,

$$\Delta J = (S \sin \theta) (\omega_0 \Delta t), \quad (2)$$

where ω_0 is the frequency of the precession, known as the Larmor frequency.

Note here, there is no orbital motion so $J = S$.

The rate of change of the angular momentum is then

$$\frac{dJ}{dt} = \omega_0 S \sin \theta. \quad (3)$$

This rate of change must be equal to the applied torque so, substituting in Equation 1

$$\begin{aligned}
\tau &= \mu B \sin \theta = \omega_0 S \sin \theta \\
\omega_0 &= \frac{\mu}{J} B \\
&= \gamma B.
\end{aligned} \tag{4}$$

2.2 Rotating frame of reference

We can transform our reference frame to one that is rotating at ω about $\mathbf{B}$. We consider the magnetic moment as a vector in our semi-classical picture, that is,

$$\vec{\mu} = \mu_x \hat{\mathbf{x}} + \mu_y \hat{\mathbf{y}} + \mu_z \hat{\mathbf{z}}. \tag{5}$$

The precessing vector then would have rate of change,

$$\begin{aligned}
\frac{d\mu}{dt} &= \frac{\partial \mu_x}{\partial t} + \mu_x \frac{d\hat{\mathbf{x}}}{dt} + \frac{\partial \mu_y}{\partial t} + \mu_y \frac{d\hat{\mathbf{y}}}{dt} + \frac{\partial \mu_z}{\partial t} + \mu_z \frac{d\hat{\mathbf{z}}}{dt} \\
&= \frac{\partial \mu}{\partial t} + \omega \times \mu \\
\frac{\partial \mu}{\partial t} &= \gamma \mu \times \left[\mathbf{B} - \frac{\omega}{\gamma} \right] \\
&= 0 \quad \text{for } \omega = \omega_0 = \gamma \mathbf{B}.
\end{aligned} \tag{6}$$

We can see that if the rotating frame is rotating at the same speed as the Larmor frequency, the magnetic moment μ , will appear stationary, hence resonance.

2.3 Magnetisation

Rather than just one magnetic moment, or one nucleus, we need to consider the whole magnetisation of the material, M ,

$$\mathbf{M}_0 = \left(\frac{\chi_0}{\mu_0} \right) \mathbf{B}. \tag{7}$$

The static magnetisation $\mathbf{M}_0$ is the vector sum of all the magnetic moments from the individual nuclei. There should be a slight excess of nuclei with components of magnetic moments parallel to $\mathbf{B}$ rather than anti-parallel, so there should be a net magnetisation $\mathbf{M}_0$ parallel to the magnetic field. In equilibrium, there will be no component of magnetisation perpendicular to the magnetic field.

If the magnetisation is perturbed, such that the magnetisation and the magnetic field are no longer aligned, the magnetisation will precess about the magnetic field,

$$\frac{d\mathbf{M}}{dt} = \gamma \mathbf{M} \times \mathbf{B}. \tag{8}$$

This precessing magnetisation can be detected if a coil was wrapped around the material.

2.4 Effect of Radio Frequency Pulses

We need to first label our static magnetic field B_0 , and place it along the z-axis, for simplicity sake, $\mathbf{B} = B_0\hat{\mathbf{z}}$. We want to now apply a plane-polarised, oscillating RF field of the form,

$$\mathbf{B}_1(t) = 2B_1 \cos \omega t \hat{\mathbf{x}}. \quad (9)$$

This pulse can be decomposed into two counter-rotating components, where one component rotates opposite to the precessing magnetisation and because it is so far off resonance, can be ignored (averages out to zero). If the frequency of the pulse is close to resonance, that is $\omega \approx \omega_0$, the rotating field will appear static in the rotating frame. Therefore, in this rotating frame, the magnetisation only 'sees' the static vector $\mathbf{B}_1$. The magnetisation vector in the rotating frame will now begin to precess about $\mathbf{B}_1$ with angular frequency, $\omega_1 = \gamma B_1$,

$$\frac{\partial \mathbf{M}}{\partial t} = \gamma \mathbf{M} \times \mathbf{B}_1. \quad (10)$$

If the rotating field is applied in the form of a pulse of duration τ_p , then the angle of precession due to the rotating field will be

$$\begin{aligned} \theta_p &= \omega_1 \tau_p \\ &= \gamma B_1 \tau_p. \end{aligned} \quad (11)$$

We can then select θ_p to change the direction of magnetisation by applying different pulses.

2.5 Bloch Equations

We can see that in Equation 8, that it lacks any effects for the relaxation of the magnetisation after the perturbation. Bloch added terms to this equation based on two conditions

1. Recovery of z component of magnetisation M_z back to its equilibrium value M_0 (in absence of perturbing RF field) with time constant T_1 , representing the longitudinal or spin-lattice relaxation time. This time relates to the magnetisation exchanging energy back into the environment to achieve thermal equilibrium
2. Decay of the transverse components of magnetisation M_x and M_y over a time scale, T_2 , characterised by the spin-spin relaxation time. Note that, in practice, the applied magnetic field B_0 is not homogenous so spins in different parts of the material will precess at slightly different frequencies. Following a 90° pulse, the magnetisation will de-phase, which has a time scale much shorter than T_2 so $T_2^* < T_2$. The slightly different local effective fields result in a broadening or splitting of NMR spectrum into discrete resonance peaks. The FWHM of the peak is given by $\frac{1}{\pi} T_2^*$.

The Bloch equations are derived from the differential equations,

$$\begin{aligned}
\frac{dM_x}{dt} &= -M_y\omega_0 - \frac{M_x}{T_2} \\
\frac{dM_y}{dt} &= M_x\omega_0 - \frac{M_y}{T_2} \\
\frac{dM_z}{dt} &= \frac{(M_0 - M_z)}{T_1}
\end{aligned} \tag{12}$$

which have solutions

$$\begin{aligned}
M_x(t) &= [M_x(0) \cos \omega_0 t + M_y(0) \sin \omega_0 t] e^{-t/T_2} \\
M_y(t) &= [M_x(0) \sin \omega_0 t - M_y(0) \cos \omega_0 t] e^{-t/T_2} \\
M_z(t) &= [M_z(0) - M_0] e^{-t/T_1} + M_0.
\end{aligned} \tag{13}$$

3 Experimental Setup

3.1 Finding FID signal

After 90° pulse, the magnetisation will precess around B_0 at ω_0 . Initially we have

$$M_z = 0 \tag{14}$$

$$M_\perp = \sqrt{M_x^2 + M_y^2} = M_0 \tag{15}$$

We know that the perpendicular magnetisation evolves with time via an exponential decay,

$$M_\perp = M_0 e^{-\frac{t}{T_2}} e^{-i\omega_0 t}. \tag{16}$$

Now we can use Faraday's Law,

$$V(t) = -\frac{d\phi}{dt} \tag{17}$$

where $\phi = \int B_\perp(t) \cdot dA$. We can then say that

$$\begin{aligned}
V(t) &\propto -\frac{dM_\perp}{dt} \\
&\propto M_0 \left(\frac{1}{T_2} + i\omega_0 \right) e^{-\frac{t}{T_2}} e^{-i\omega_0 t} \\
&\propto e^{-\frac{t}{T_2}} e^{-i\omega_0 t}
\end{aligned} \tag{18}$$

3.2 Different samples

In the case of solid materials and viscous liquids, with short intrinsic transverse relaxation times T_2 , the spin-spin relaxation function can be determined directly from the FID or spectrum. However for non-viscous liquids, where the FID signal is affected by the magnetic field inhomogeneities, leading to a transverse relaxation time characterised by T_2^* .

Nuclei in diamagnetic samples will be partially screened from the full effect of the applied static magnetic field, representing a 'chemical shield'. This shield arises from the electron cloud of the atom, once the magnetic field is applied, it will induce a counter magnetic field, reducing the effect of the applied magnetic field. The inhomogeneity of the sample is then reflected in the differences in precession frequencies distributed in the spins in the sample. For liquids, this screening is isotropic (as the motion of the atoms are random). The field then experienced by a nucleus is then reduced to

$$B = B_0(1 - \sigma) \quad (19)$$

where σ is a dimensionless, chemical shielding constant. The corresponding precession frequency is then

$$\omega = \gamma(1 - \sigma)B_0 \quad (20)$$

and the difference in average precession for two groups of nuclei (of the same atomic species but in different samples) is then

$$\Delta\omega = \gamma(\sigma_2 - \sigma_1)B_0. \quad (21)$$

The chemical shift, δ , is then calculated as a fraction of the reference field/frequency,

$$\delta = \left(\frac{\Delta\omega}{\omega_{\text{ref}}} \right) \times 10^6, \quad (22)$$

given in ppm.

3.3 Measuring the characteristic times

To measure T_1 , we use a pulse sequence by first perturbing the magnetisation (aligned with the z-axis), with either a 90° or 180° pulse, then letting the magnetisation recover over time τ . After τ , the magnetisation is 'read out' by applying a 90° pulse and recording the FID.

In the first case, where we first perturb with a 90° pulse (saturation recovery), we want to equalise the populations of states, corresponding to an infinite spin temperature state. During the relaxation delay τ , the transverse magnetisation de-phases and the longitudinal component recovers, so the FID amplitude,

$$M_z(\tau) = M_0 \left(1 - e^{-\tau/T_1} \right). \quad (23)$$

To help us obtain T_1 , we can make an approximation, in liquid-like samples and biological materials,

$$M_y(t) = M_z e^{-t/T_2^*}. \quad (24)$$

In the case of starting with a 180° pulse (inversion recovery), the sequence reverses the initial state populations and sample magnetisation, corresponding to a state of negative spin temperature. Following the 90° pulse after recovery time τ , the magnetisation recovers according to

$$M_z(\tau) = M_0 \left(1 - 2e^{-\tau/T_1} \right). \quad (25)$$

Here, M_z will pass through zero when $\tau = T_1 \ln(2)$.

So far, we have only been able to measure T_2^* . In order to measure T_2 directly, we need to employ a 'spin echo' method. Following the initial 90° pulse, we can reverse the contribution to de-phasing of the transverse magnetisation by applying a 180° pulse. The spins will then rephase momentarily before starting to dephase again.

3.4 Apparatus

The basic layout of this experiment involves a coil wrapped around a vial of the sample material. The induced voltage on the coil due to the change in the magnetisation of the sample is then analysed.

3.4.1 Receiver Module

The sample coil signal is first amplified by a low noise amplifier which has a fixed gain of 20 dB. This signal was subsequently fed into a variable gain amplifier, with a range of 0-80 dB, but we chose to set the gain to 75 dB. The goal of amplifying the signal, which we will discuss again with the capacitors, is to maximise signal but to a point before the signal is saturated, due to instrument constraints. The signal is filtered by a bandwidth filter, specifically designed for either hydrogen or fluorine nuclei detection. The output of the filter is sent to the inputs of the envelope detector and phase sensitive detectors.

3.4.2 Synthesiser

The objective of the synthesiser is to tune the experimental setup to the Larmor Frequency as to satisfy the condition found in Equation 6. One of the nice consequences of achieving resonance is that we are able to obtain the pure FID found in the voltage that we are measuring in Equation 18. To do so, we need to demodulate the induced voltage by multiplying our synthesised continuous wave by the signal. If we consider the real component of the voltage,

$$V_{\text{real}}(t) = e^{-\frac{t}{T_2^*}} \cos(\omega_0 t), \quad (26)$$

and we multiply this signal by a continuous wave,

$$\begin{aligned} V_{\text{sig}}(t) \cdot V_{\text{ref}}(t) &= e^{-\frac{t}{T_2^*}} \cos(\omega_0 t) \cos(\omega_{\text{ref}} t) \\ &= \frac{1}{2} e^{-\frac{t}{T_2^*}} [\cos(\omega_0 - \omega_{\text{ref}})t + \cos(\omega_0 + \omega_{\text{ref}})t]. \end{aligned}$$

The second term will have a very high frequency oscillation and so can be easily filtered out. If we just consider our beating term, we can achieve resonance when this term is constant, when $\omega_{\text{ref}} = \omega_0$. We can then read the voltage from an oscilloscope which will just return us the FID.

3.4.3 The Permanent Magnet and LC Circuit

The permanent magnet was used to apply the homogenous magnetic field B_0 along the z-axis of the sample coil. There were 4 shim coils to adjust the magnetic field gradient as to improve the homogeneity of the field. Optimising these coils were relatively easy as the coils' optimal position was when the tail of the FID curve was maximised.

The permanent magnet was made out of NdFeB or Neodymium. This material is quite susceptible to temperature change. On the e-magnetsuk.com website, it cites that for a M type magnet (max working temperature of 80°C), the temperature coefficient is -0.12, which represents a 0.12% change in the magnetic field strength for every 1° temperature change. The permanent magnet is encased in a temperature cooling system, designed to keep the magnetic poles at a stable temperature however the manufacturer explicitly says that this system is only stable if the room temperature is kept to a $\pm 1^\circ\text{C}$ stability.

The manual also makes an explicit note that the magnetic drift error is on the order of the difference from the Larmor frequency of hydrogen vs fluorine. The precession frequencies are only 6% different and the permanent magnet will vary by several percent on different units at ambient room temperature. The marked frequency on the magnet is merely a good place to start "tuning up" the spectrometer.

Finally, the sample inductance coil is a part of the LC circuit set up to maximise the signal obtained by the receiver module. The capacitor is then most importantly needed to tune the circuit to the correct driving frequency, the Larmor frequency. We found that our capacitor could only be tuned by rotating the screw to change the positions of the plates.

3.5 Uncertainty Analysis

The most challenging component of this experiment will be tuning the circuits to resonant frequency of the precessing magnetisation i.e finding the Larmor frequency of the sample. The desired frequency depends on the magnetic field strength which, according to the manual has several percentages error, on order of 6% around the expected frequency of 21.7 MHz.

Another large source of systematic error will be the position of the sample in the coil. The solenoid itself is 30 mm long with only one half it being 'reasonably' uniform. Our sample's vertical height was approximately 3 mm. Combined with the 'optimal' tuning of the magnetic field gradient coils, the homogeneity of the magnetic field is estimated to be 1% error. The manual maintains that the magnetic field maintains homogeneity effectively, even untuned.

Furthermore, the tuning of the capacitors would contribute to bias. With some signal reflected back in an off-resonant circuit, this would only affect the strength of the signal received, not necessarily finding resonant frequency. The capacitors were tuned using a screw to change the position of the plates. We found that there were 10 full revolutions for the range of the capacitor so our error should be half a rotation, or 5% error.

Lastly, the temperature of the room would drastically affect the bias in the experiment. We did have the air conditioning unit on during the whole day (9 am to 6 pm) however the temperature outside did change drastically that day. On September 24th, 2025, the Bureau of Meteorology recorded an average temperature across Sydney to be 16.8°C at 9 am and then 23.9°C at 3 pm, with a max of 26.1°C during midday. This clearly violates the stability condition stated by the manufacturer of $\pm 1^\circ\text{C}$, as a conservative estimate of the change of temperature in the room would be around that range. We can estimate the percentage on this instability error to be 1% as the sample was largely confined within a temperature controlled environment inside the magnet.

The total systematic error is then

$$\sigma_{\text{sys}} = \sqrt{0.06^2 + 0.01^2 + 0.05^2 + 0.01^2} = 0.08 = 8\%. \quad (27)$$

Without any signal, the oscilloscope measured an oscillating background noise of about ± 10 mV. This would encompass electronic and thermal noise. In terms of random fluctuation in the signal, they would be dominated by thermal motion of spins and quantum fluctuations. As there is a large number of spins, on scale much greater than 10^{10} , we can assume the spin noise to be Gaussian and its standard deviation to scale with $\sqrt{N}$. Therefore, the percentage error would be $\frac{1}{\sqrt{N}}$, an incredibly small number so we can neglect these effects. Lastly, the greatest contribution to statistical error would be instrument resolution. We need to account for oscilloscope trigger, pulse length resolution, sample rate which would effect the sharpness of FID peaks. The largest of the resolution errors, would be the pulse duration resolution. We are going up in $0.5 \mu\text{s}$ so the statistical error contribution would be $\pm 0.25 \mu\text{s}$.

4 Experimental Results

Rather than describing the experiments, we will focus more on what we did and what went wrong. Firstly, we successfully obtained a value for T_2^* by producing a FID seen in Figure 1 below. Fitting the curve to the data, our estimated value for T_2^* is 3.51 ± 0.02 .

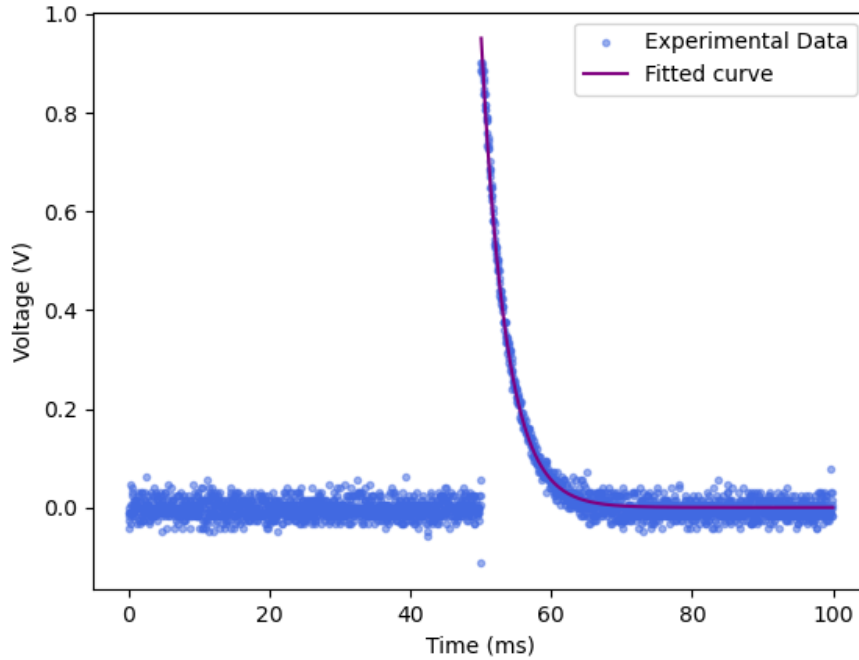


Figure 1: This is a voltage vs time graph clearly displaying a FID curve with a sharp peak. We find that because of our low signal, the 10 mV noise is apparent in the plot. This is reflected in the statistical error obtained by the fitting curve. The standard error for the amplitude parameter was 8% and the error for T_2^* was 0.6%. We find that the slope is quite accurately fitted whereas the amplitude has more variation, which aligns with the visual from the graph.

However, the manual for this experiment expects our value for T_2^* to be greater than 5 and so we realised

our set up for this FID was not optimal, i.e our signal strength was too weak. The reason for this being the time of decay of the transverse magnetisation will increase if we had more initial spins magnetised in the direction of our magnet. We had not realised we needed to increase the signal at this time, so we proceeded with the subsequent experiment.

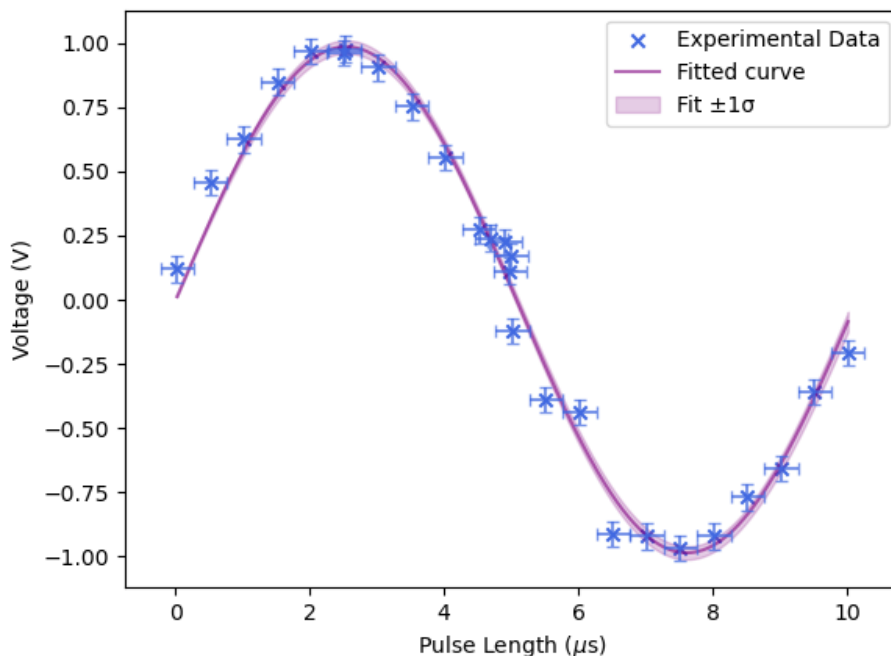


Figure 2: The peak FID amplitude displays a sinusoidal relationship with the duration of the pulse. We find that the signal is particularly weak as near the points of interest, i.e the peak and troughs of the graph and the point at which the amplitude is zero, we find more varying behaviour. We find that the standard error in the frequency of the graph is 6% using the curve fit parameters. The extrema of the graph occur when the pulse length is $2.52 \mu\text{s}$ and $7.59 \mu\text{s}$.

Figure 2 shows a clear sinusoidal relationship. It had taken us many hours to get to this point because we could always tell the synthesiser was not on resonance if the FID amplitude would not go to zero near $5 \mu\text{s}$. Most of the time, when we were tuning the synthesiser, the FID amplitude would go near zero, but never zero as we were off resonance and so there was always inherent changing magnetisation, regardless of pulse length, as per Equation 6. Once we achieved this sine curve, or at least a curve that supposedly went to zero, we were unsure as to why it was not as smooth as a sine curve. The extrema points obtained were 2.52 ± 0.15 and $7.59 \pm 0.46 \mu\text{s}$, which agree with expected results given in the manual. However, the large standard errors were a worry. One of the things we checked was the scale of the oscilloscope and we realised that we were not exactly at zero throughout the sweep. Therefore, we were not at resonance and so the search for resonance started again. We increased the signal and the driving frequency and then obtained the same plot, with 10 times the original signal.

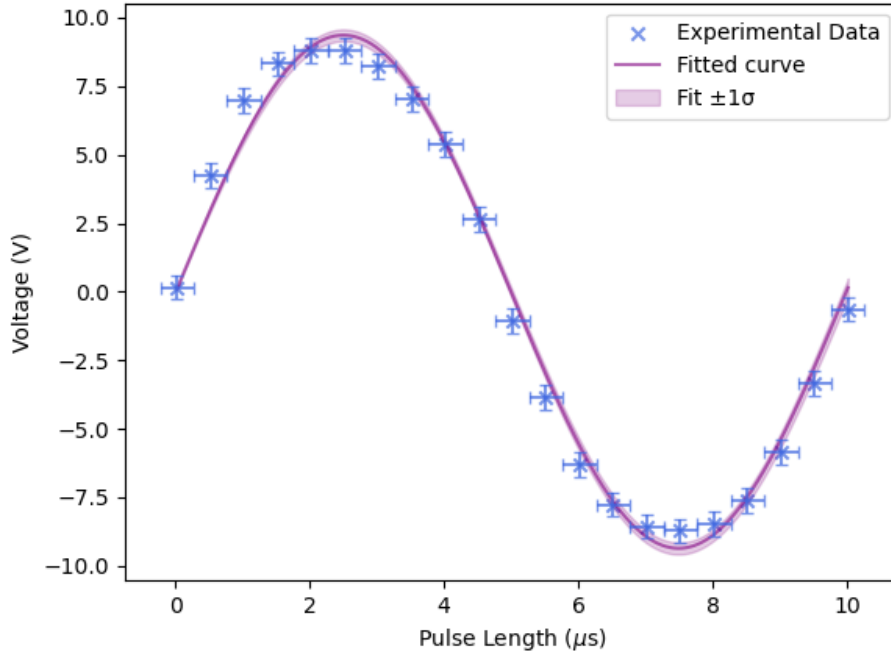


Figure 3: Same plot as before but with 10x the signal. We find a smoother relationship visually. We find our standard error for the frequency of the graph goes down to 5%. The behaviour around the extrema are less chaotic due to higher signal to noise. The corresponding pulse length values are 2.48, 5.02 and 7.47 μs for 90° , 180° and 270° respectively.

There is clearly a smoother relationship between the peak amplitudes and the pulse length and so this was a much more certain result for the sweep. We found the peak amplitude for the FID actually went to 0 at $5 \pm 0.25 \mu\text{s}$ and so were satisfied with the result. However, we then checked the FID decay curve for the 90° pulse length. Upon closer inspection we found the curve shown in Figure ??.

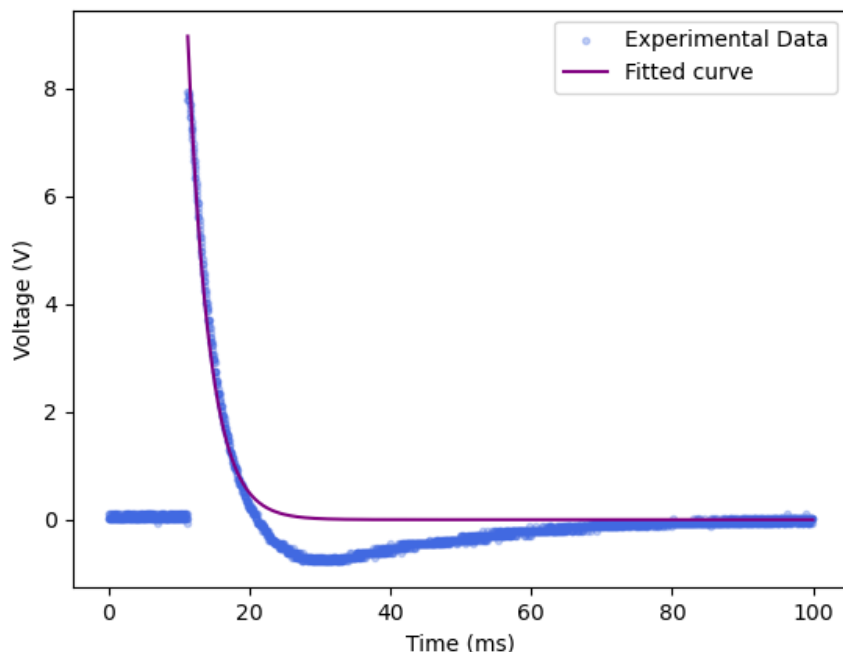


Figure 4: Comparing with the previous FID curve, we find this to be 10x the signal again. However this time, there is a large dip right after the decay of the curve. This does not impact the FID curve directly however pulses after this initial curve would not be read correctly.

There is a large, non-negligible dip after the FID curve, and the FID curve itself is incredibly short. Despite the increase in signal to noise, we found that our T_2^* value decreased drastically to $3.02 \pm 0.04 \mu\text{s}$, suggesting that the curve itself has been 'eaten' up by the dip. We believe that the increase in signal had caused our receiver to not recover in time, or saturate.

5 Discussion

Since there really is not an overarching story here, or path of discovery about pulsed nuclear magnetic resonance, I will use this section to dot point our pain points and thoughts. The lab book in appendix will detail our thoughts on the day. We completed this lab over 2 days, in the first day, we had very little knowledge of what the lab entailed, outside of the theory. On the second day, after reading the manual, we had a good understanding of the instruments and apparatus used.

1. The first thing we noticed going into the lab on the second day was that our sample volume was much larger than suggested. We had not picked this up on the first day hence why we believe things did not work on the first day. It is important that the sample lies within the effects of the sample coil as too much sample would lead to spins resisting magnetisation, potentially introducing unnecessarily friction.
2. Over the week, we realised that the only important thing in this experiment was to get the synthesiser to resonant frequency. It is important to optimise all components of the equipment however in terms of being able to achieve results, especially in the second experiment, we needed to drive the setup at the Larmor frequency. It seems simple but this one condition became the bane of the experiment.

3. Once we had decreased the volume of the sample, we changed the synthesiser frequency to 21.3 MHz, as this maximised the amplitude of the signal. However, this would not be resonant frequency as the FID curve would not go to zero when the pulse duration was supposedly at 180° . All we did was tune the synthesiser frequency to the frequency of the capacitors.
4. The capacitors did not play a role in finding resonant frequency, they served the exact same function as the variable gain amplifier, all they did was maximise signal. We naively spent so much wasted time tuning the capacitors before achieving resonant frequency of the synthesiser and so we managed to tune them to the same frequency but it was not the Larmor frequency.
5. We realised that we could use one of the outputs of the synthesiser to help us find resonant frequency. The synthesiser demodulates the signal by multiplying it by the reference signal that we choose. We achieve resonant frequency when the beating between these two frequencies is 0 which is given by one of the outputs from the synthesiser. The synthesiser was then tuned to when this beat was zero and finally we had achieved a sine curve, or atleast the FID amplitude going to zero. However, we realised soon that our oscilloscope's scale was too high and so it was not actually resonant frequency. At this point in time, we tuned it to 21.87 MHz.
6. The reason we were unsure of our previous frequency was because our T_2^* could never get close to 5, it was always 4.26 and below no matter how optimal our magnetic field gradients were. This was another pain point, the magnetic field gradients were so difficult to use and there was no clear direction in how they should be tuned. The manual provided little help here.
7. We finally tuned our synthesiser to 21.89 MHz and found a perfect sine curve shown in experimental results. Now, we ran into the problem of having not enough signal for Experiment 3. The second pulses were way too small for a good reading and so we started to play with the capacitors again. We tuned the capacitors to the point where we could achieve 40 peak to peak voltage however, once we looked at the FID curve, we found that the FID curve saw a large dip after the decay, hinting at instrument saturation.
8. After this time, it had been a couple of hours since we achieved the sine curve, and with so much time passing, our set up was now out of resonance. We could not replicate that sine curve again. The last hours in this experiment were spent trying to achieve the resonance we once had, even trying the old methods however we could not get it again.

6 Appendix

6.1 Pre-lab questions

6.1.1 Question 1

The Larmor angular frequency is given as

$$\omega_0 = 2\pi f_0 \quad (28)$$

We can substitute this into Equation 4, rearranging for B_0 ,

$$\begin{aligned} B_0 &= \frac{\omega_0}{\gamma} \\ &= \frac{2\pi(21.7 \times 10^6)}{2.675 \times 10^8} \\ &= 0.51 \text{ T.} \end{aligned}$$

6.1.2 Question 2

The gyromagnetic ratio of ^{19}F nuclei is given as

$$\gamma_{19} = 2.51815 \times 10^8 \text{ s}^{-1}\text{T}^{-1}.$$

Therefore the Larmor Frequency, using Equation 4 in the field in Question 1 is

$$\begin{aligned}\omega_0 &= \gamma_{19}B_0 \\ &= 2.51815 \times 10^8 \times 0.5097 \\ &= 128.4 \text{ MHz}\end{aligned}$$

or equivalently,

$$f_0 = \frac{\omega_0}{2\pi} = 20.4 \text{ MHz} \quad (29)$$

6.1.3 Question 3

The beat frequency is the difference of the two frequencies so

$$f_{\text{beat}} = 21.7 - 21.696 = 0.004 \text{ MHz}. \quad (30)$$

The oscillation period is simply the inverse,

$$T = \frac{1}{0.004 \times 10^6} = 0.00025 \text{ s} = 0.25 \text{ ms} \quad (31)$$

6.1.4 Question 4

We use Equation 22, with the reference frequency being 20.0000 MHz. The chemical shift is

$$\begin{aligned}\delta &= \frac{\Delta\omega}{\omega_{\text{ref}}} \\ &= \frac{20.0001 - 20.0000}{20.0000} \times 10^6 \\ &= 5 \text{ ppm}.\end{aligned}$$

6.2 Lab Book

Operating instructions

PS-2 spectrometer

- Can only study NMR signal from hydrogen nuclei and fluorine
- RF probe head has single coil and 50Ω impedance

Receiver module

- Amplify small voltage induced in sample coil to a large enough amplitude for the oscilloscope
- Signal is amplified by low noise amplifier (LNA)

LNA

- has fixed gain of 20dB (100x)
- noise figure of 2.5 dB = $10^{\frac{2.5}{10}} \approx 1.78 \times$ worse SNR
- Output of LNA is fed into variable gain amplifier (0-80dB)
- then signal goes through a Bandwidth filter (p for proton, f for fluorine)
 - ↳ buffered version now available at RF out connector → Clean sine wave at Larmor Frequency
- Output of filter is sent to inputs of envelope detector and phase sensitive detectors
 - ↳ Envelope detector tracks amplitude of RF voltage → sends to Env. Out
 - ↳ phase sensitive detector is product of Ref In signal and filter output → I out
 - ↳ Q out signal is Ref In signal shifted 90° and filter output
- by adjusting phase of RF out signal, we make these signals correspond to in-phase and quadrature components of NMR signal

$V(t)\cos\omega t + iV(t)\sin\omega t \rightarrow (I + jQ)$
 Fourier transform to get spectrum

ASSUMPTION: Receiver coil is tuned to match 50Ω of coaxial cable connecting sample coil to connector on receiver module (named SAMPLE COIL)

- We need to make sure pulsed RF output of synthesiser is coupled to sample coil (ensure B_1 rotates the direction of spin polarisation)
- during RF pulse, the input in receiver cannot interfere with RF pulse to sample coil and after RF pulse input of receiver must be coupled to sample coil so to amplify signal but needs to disconnect from output of synthesiser.

↳ use diodes

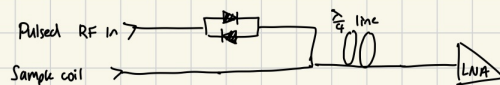
Diodes used are silicon p-n junctions with I-V characteristics:



$$I = I_0 e^{\frac{V}{V_0}} \quad \begin{array}{l} \text{induced voltages} < V_0 \\ \text{typically} \end{array}$$

- during RF pulses RF output of synthesiser swings $\pm 25V$ (alternating +25V and -25V relative to g_m)

• The $\frac{\lambda}{4}$ transmission line makes everything work



↳ transforms impedances on both ends $Z_{in} = \frac{Z_0^2}{Z_{load}}$

↳ If diodes conducting (Pulse transmit to sample coil), diode is open (0Ω), input to $\frac{\lambda}{4}$ line is a short circuit so LNA is disconnected. vice versa

• Pulse programmer outputs a blanking signal, this reduces gain of RF amplifiers during RF pulse from overloading and prevent the capacitors in outputs of detectors from being charged.

↳ ensures recovery of circuit after pulse

↳ change interval using Blanking duration, account for different sample coil tunings

Synthesiser

• proton and fluorine precession are only 6% different → field of permanent magnet varies unit to unit by several percent and will drift with ambient temperature

↳ uncertainty in frequency from magnet variation and drift is on same order as H-F separation

↳ cannot rely on a synthesiser setting to distinguish protons from fluorine

↳ need to tune magnet to actual resonance frequency

↳ manual suggests start synthesiser frequency at marked precession frequency (proton at ambient room temp) and then tune up the spectrometer (adjust RF freq)

For pulse operation, we only care about

• REF OUT (REF IN for receiver module)

↳ provides continuous wave, sinusoidal signal to phase sensitive detectors in receiver

↳ basically 'washes' out $\sim 21\text{MHz}$ proton precession frequency, the mix remains the lower frequency decay curve

↳ Basically, after sample coil, the voltage signal is an amplitude modulated wave

$$V(t) = A(t) \cos(\omega_0 t + \phi)$$

$\uparrow$ FID decay $\uparrow$ Larmor Frequency $\uparrow$ Starting position (relative to $z = z_{max}$ is max)
 $\phi = 0$ is max

the CW will wash away the carrier wave, demodulation

• Pulsed RF Out: high power RF signal sent to coil surrounding sample via receiver module

- Pulse In, I, Q : Corresponds to I, Q, pulse out connectors in pulse programmer module
↳ Shapes RF output power used to 'tip' nuclear spins

Pulse Programmer

- adjust pulse length to achieve 90° and 180° rotation of magnetisation
- options are self explanatory
- Toggle M4 disables Meibohm-Gill 90° phase shift between A and B, measures T_2 with multiple pulse B's
- Blanking out pulse slightly longer than A and B, disables receiver during pulse, prevents internal capacitors from charging

The Permanent Magnet

- NdFeB material has its magnetisation dependent on temperature
- Just make sure light is off!
- Shim coils fixed inhomogeneity in different directions: z^2 is parabolic correction

RF sample probe head

- Tuning capacitors (left), the two of them combine to tune sample coil into series resonance at frequency of applied RF signal
↳ Set the frequency in the LC circuit, needs to be at resonance so maximum current flows $\rightarrow$ max signal
- Matching capacitor makes coil impedance 50Ω so power transfer between RF amplifier and LNA on receive is efficient $\rightarrow$ if not 50Ω we see lower amplitude
↳ fixed matching capacitor has 400 pF $\rightarrow$ always atleast 976 pF sample probe is usually well enough matched (look back at assumption)
- the inductance coil has to be on resonance with RF frequency of precession frequency of nuclei
↳ tune to frequency of synthesiser

Other notes

- volume of sample heavily influences results
↳ the solenoid is 30mm long, RF is only reasonably uniform about half of its overall length
- If we can't achieve 40 V peak to peak something is broken
- creation of transient x-y magnetisation on time scale of 10^{-6} s but precession on time scale 10^{-8} s
- I/Q output is product of signal from precessing spins and reference signal

$$\sin(\omega_{\text{ref}} t) \cdot \sin(\omega_{\text{spin}} t) = \frac{1}{2} \underbrace{\cos(\omega_{\text{ref}} - \omega_{\text{spin}}) t}_{\text{beat is 0 at resonance}} - \frac{1}{2} \underbrace{\cos(\omega_{\text{ref}} + \omega_{\text{spin}}) t}_{\text{should be filtered out because too large}}$$

- Care for aliasing: zero beat signal due to sampling pattern of oscilloscope - Change from sampling mode to peak detection

In-lab notes

- Sample still off resonance
- Our sample liquid was way higher volume than suggested 5mm
- We decreased the volume of sample, changed synthesiser frequency to 21.3 MHz as it provided max amplitude of signal
- We finally obtained a one peak sweep i.e. a sine curve!
 - ↳ We changed the synthesiser frequency to 21.87 MHz when the beat frequency went to zero.
- Upon doing the first experiment, we found we could not obtain a T_2^* greater than 5 as suggested by manual (we got 4.26)
- Oops, we found that our scale for I/Q was too large so we thought it was zero but wasn't → make scale smaller turn syn to 21.89 MHz
- At experiment 3 we found that we couldn't get a strong enough signal for 2nd peak so we tuned the capacitors again - the calibration was completed by achieving a 40V ptp voltage with the probe
- After tuning capacitors we see a weird dip after the FID for pulse A
 - ↳ the dip goes away if pulse A goes away
 - ↳ receiver saturation

Exp 3

Syn: 21.89934 MHz

90° : 2.5 μ s

180° : 4.8 μ s